

## Homework Stochastic Simulation - Fourth set

For the programming exercises, you are free to use any programming language or mathematical software package (Maple, Matlab, Mathematica, R, anything) as long as you

- include all code in your deliverables, and
- for random number generation purposes, you use standard built-in methods to generate standard uniform samples. For generating samples from any other distribution, you'll have to construct them yourself using the standard uniform samples.

You are expected to submit the homework individually. Please compile answers and code in a single PDF document and upload it to canvas. For questions, you can contact Tygo Nijsten at [t.g.c.nijsten@uva.nl](mailto:t.g.c.nijsten@uva.nl).

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**Exercise 1** Let  $X$  be a random variable which is Weibull distributed with parameters 2 and  $\lambda$ . In other words, its density function is given by

$$f_X(x) = 2\lambda^2 x e^{-\lambda^2 x^2} \mathbb{1}_{\{x>0\}}.$$

Furthermore, let  $Z = h(X)$ , where  $h(x) = 2^x$ .

- Explain briefly how one can estimate  $\frac{d}{d\lambda} \mathbb{E}[Z] \big|_{\lambda=\frac{1}{2}}$  using the method of finite differences, while not using common random numbers.
- Explain briefly how one can estimate  $\frac{d}{d\lambda} \mathbb{E}[Z] \big|_{\lambda=\frac{1}{2}}$  using the method of finite differences and using common random numbers.
- Explain briefly how one can estimate  $\frac{d}{d\lambda} \mathbb{E}[Z] \big|_{\lambda=\frac{1}{2}}$  using the method of infinitesimal perturbation analysis.
- Explain briefly how one can estimate  $\frac{d}{d\lambda} \mathbb{E}[Z] \big|_{\lambda=\frac{1}{2}}$  using the likelihood ratio method.
- Implement each of these methods in a simulation program. Then, compare how well each of these methods works in this case (e.g. by computing sample mean squared errors). How do the methods rank in order of estimator quality? Report and argue your findings.

**Exercise 2** In this exercise, we consider the optimization problem of finding

$$\min_{\theta \in \Theta} \mathbb{P}(\max\{\theta X_1 + X_2, (1 - \theta)X_3\} \geq 3),$$

where  $\Theta = [0, 1]$ ,  $X_1 \sim \text{Gamma}(2, 2)$ ,  $X_2 \sim \text{Gamma}(2, 1)$  and  $X_3 \sim \text{Gamma}(2, \frac{2}{3})$ . You may assume that the probability is smallest for some  $\theta$  in the interior of  $\Theta$ : the probability is not minimized when  $\theta = 0$  or  $\theta = 1$ . We estimate the solution by using the iteration  $\theta_{n+1} = \Pi_{\Theta}(\theta_n - \frac{C}{n+1} D(\theta_n))$ , where  $D(\theta_n)$  is a derivative estimate at  $\theta = \theta_n$  for some value of  $C$ .

- When using the likelihood ratio method for the derivative estimates with  $R = 10$  and taking  $C = 0.2$  and  $\theta_0 = 0.5$ , plot a graph of the  $(\theta_n)_{n \geq 0}$ -sequence as a function of  $n$  for  $n = 0, \dots, 50000$ . What is the found solution to the above optimization problem?
- Experiment with other values of  $C$ : does one get better results? Make sure to include in your answer a description of the tradeoff that must be struck when finding the 'right' value of  $C$ .